# Betelhem Asefa

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## EDUCATION

**Towson University** **Towson, MD**
*Bachelor of Science in Information Technology* *May 2026*
Concentration: Data Analytics and Management | Minor: Business Analytics
**Availability:** Graduated May 2026; available to start November 2026
**Relevant Coursework:** Business Statistics, Database Management, Financial Analytics, Project Management, Excel, Tableau

## SKILLS

**Data & Analytics:** SQL (MySQL: CTEs, window functions, views, stored procedures, triggers), R (tidyverse, caret), Python (pandas), Excel
**Visualization:** Tableau, ggplot2, matplotlib, seaborn
**Methods:** Customer segmentation (K-Means), classification modeling, text analytics (TF-IDF, sentiment analysis, LDA topic modeling)
**Other:** Java, HTML, Claude (development, debugging, analysis)

## ANALYTICS PROJECTS

**Xbox Game Platform Database (MySQL)** **Fall 2025**
- Designed and built a 10-table MySQL database covering a gaming platform's players, game and DLC catalog, purchases, subscriptions, and multiplayer sessions, with keys, constraints, and cascade rules to keep data accurate
- Wrote 12 analytical SQL queries using CTEs, window functions, and CASE logic to measure revenue by title and developer, average order value, customer spend tiers, DLC attach rate, and subscription adoption by region
- Automated reporting with KPI views, summary tables, stored procedures, and functions; added triggers for purchase auditing and validation, plus role-based access for analyst vs. DBA users
- Visualized results in Python, building a revenue Pareto chart, cohort heatmap, and purchase funnel

**Customer Segmentation Using K-Means Clustering** **May 2026**
- Segmented 200 customers by age, income, and spending score using K-Means clustering in R, selecting k = 5 via the Elbow Method and silhouette analysis
- Profiled five segments and identified a high-income, low-spending group (~$86K average income) as the key growth target for promotions and personalized offers
- Delivered segment-specific marketing recommendations, including loyalty programs for high-value customers and reallocating spend away from low-value segments

**Predicting Marketing Campaign Response with Classification Models** **Apr 2026**
- Built Logistic Regression, Random Forest, and Neural Network models in R to predict customer response to a marketing campaign, using a 70/30 holdout split and 10-fold cross-validation
- Found that accuracy was misleading due to an 85/15 class imbalance; selected the Neural Network for the highest Kappa (0.508) and best balance at identifying likely responders
- Recommended targeting strategies to reduce wasted outreach to non-responders while still reaching high-value customers

**Hate Speech Detection on Social Media Using Text Analytics** **May 2026**
- Analyzed a 440,000+ post social media dataset in R, building a stratified, class-balanced sample of 6,000 posts to overcome memory limits and an 82/18 class imbalance
- Applied TF-IDF, sentiment and emotion analysis (Bing, NRC), LDA topic modeling, and n-gram analysis to identify the language patterns that distinguish hate speech
- Trained a Naive Bayes classifier and improved it by replacing sparse TF-IDF features with binary word-presence and sentiment features

## EXPERIENCE

**hellohive** **Nov 2025 – Dec 2025**
*Ambassador* *Remote*
- Built Tableau dashboards visualizing school-wide student engagement data over a 2-month period
- Partnered with student organizations and campus groups to expand event visibility and participation

## PROGRAMS & ACTIVITIES

**SEO Career – SEO EDGE Participant (Remote)** **Mar 2025 – Present**

**hellohive – helloFinance Summer Program, Selected Participant (Remote)** **Jul 2025 – Aug 2025**
- Selected from 2,000 students across 500+ schools; attended panels with Goldman Sachs, BlackRock, and Morgan Stanley professionals

**Association for Women in Science (AWIS)** – *Member* **Dec 2023 – Present**